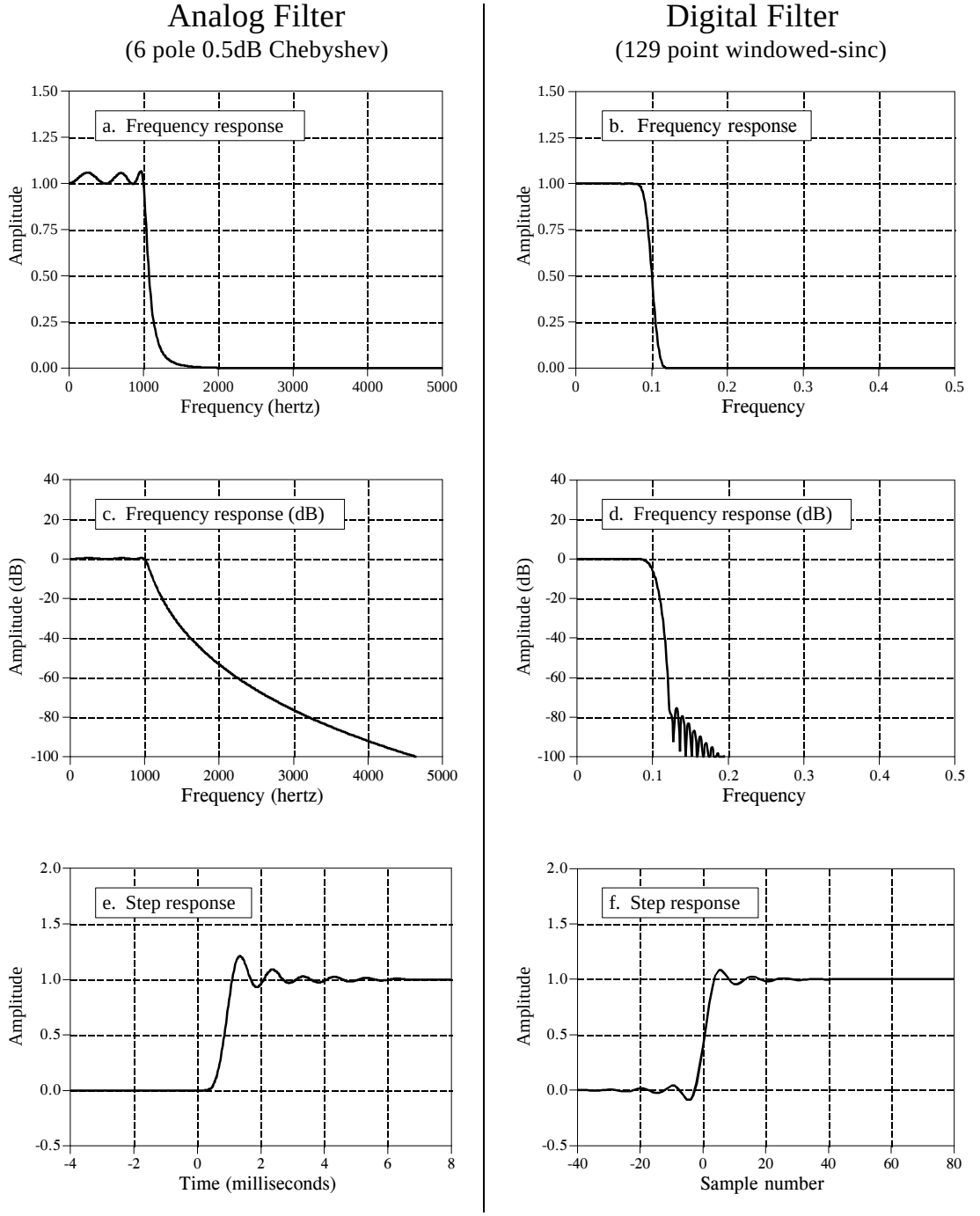

FIGURE 21-1
Comparison of analog and digital filters. Digital filters have better performance in many areas, such as: passband ripple, (a) vs. (b), roll-off and stopband attenuation, (c) vs. (d), and step response symmetry, (e) vs. (f). The digital filter in this example has a cutoff frequency of 0.1 of the 10 kHz sampling rate. This provides a fair comparison to the 1 kHz cutoff frequency of the analog filter.

a linear scale to show their exceptional filter performance, while analog filters need the logarithmic scale to show their huge dynamic range.